

ArbitrageAI Platform — Product Requirements Document (PRD)

Classification: Internal — Product & Engineering

Version: 1.0

Date: June 4, 2026

Prepared for: Technical Lead, Product Manager, Engineering Team

1. Product Purpose

ArbitrageAI is a subscription SaaS platform that enables non-technical cryptocurrency users to automatically earn profits through AI-driven arbitrage trading. Users connect their existing exchange accounts, activate the AI engine, and the system handles all strategy execution, risk management, and reporting automatically.

Primary Goal: Help users make money from crypto arbitrage without any technical knowledge.

Secondary Goal: Build a globally scalable, Africa-first platform with persistent recurring revenue.

2. Target Users

Persona 1: The African Mobile User (Nigeria / South Africa / Kenya)

- Age: 22-40 | Device: Android smartphone
- Crypto knowledge: Owns USDT or BTC for remittances or inflation hedging
- Technical level: Uses WhatsApp, mobile banking (M-Pesa, OPay, Capitec)
- Primary notification: WhatsApp
- Primary payment: Mobile money (M-Pesa, Flutterwave), bank transfer

Persona 2: The Global Retail Crypto Holder

- Age: 25-45 | Device: Desktop + iPhone
- Crypto knowledge: Holds BTC/ETH on Coinbase or Binance
- Technical level: Uses apps comfortably, not a developer
- Primary notification: Telegram or Push
- Primary payment: Credit card (Stripe)

Persona 3: The Active Crypto Trader (Pro/Institutional)

- Age: 28-50 | Understands funding rates, perpetuals, CEX/DEX
- Pain point: Manual arbitrage is exhausting and speed-limited
- Primary notification: Telegram two-way commands

3. Key Requirements

3.1 Exchange Connectivity

Req ID	Requirement	Priority
EC-01	Users connect exchanges without knowing what an API key is	P0
EC-02	OAuth 2.0 flow for Coinbase, Kraken, Luno	P0
EC-03	Guided API key wizard for Binance, Bybit, VALR	P0
EC-04	Wizard validates connection and shows balance within 2 seconds	P0
EC-05	Wizard explicitly instructs disabling Withdraw permissions	P0
EC-06	Human-readable error messages — no technical jargon	P0
EC-07	Support 2 exchanges (Starter), 10 (Pro), 20 (Institutional)	P0
EC-08	SARS AIT pin wizard for South African users	P1
EC-09	BVN/NIN KYC verification for Nigerian users	P1

3.2 Strategy Execution

Req ID	Requirement	Priority
SE-01	Funding rate arbitrage automated (Starter+)	P0
SE-02	Spatial arbitrage (Pro+)	P0
SE-03	Africa P2P premium trading (Africa Pro+)	P0
SE-04	No trade executes unless net profit > 0.15% after all fees	P0
SE-05	Every strategy passes backtesting gates before live deployment	P0
SE-06	Partial fills handled: unhedged legs never left open	P0
SE-07	Statistical/pairs arbitrage (Pro+ Phase 4)	P1
SE-08	CEX-DEX arbitrage (Institutional Phase 5)	P2

3.3 Risk Management

Req ID	Requirement	Priority
RM-01	Per-user daily loss limit auto-pause	P0
RM-02	Max capital per trade limits enforced per tier	P0
RM-03	Slippage guard: reject order if actual slippage > threshold	P0
RM-04	Platform-wide KILL_SWITCH pauses all strategies globally	P0
RM-05	Post-trade execution quality reconciliation	P0
RM-06	Strategy auto-deprioritization after 7-day underperformance	P1

3.4 Dashboard & Reporting

Req ID	Requirement	Priority
DR-01	Real-time P&L in user's local currency	P0
DR-02	FX rate refreshed every 60 seconds	P0
DR-03	FX rate stored at execution timestamp for tax reporting	P0
DR-04	SARS-compliant tax export CSV (South Africa)	P1
DR-05	FIRS format export (Nigeria, Phase 3)	P1
DR-06	Per-strategy P&L breakdown	P1
DR-07	Trade-by-trade log with full detail	P1
DR-08	Nigeria: three-rate FX display (CBN, Wise, P2P)	P1

3.5 Notifications

Req ID	Requirement	Priority
NT-01	WhatsApp Business API alerts for Africa users	P0
NT-02	Telegram Bot alerts for global users	P0
NT-03	SMS fallback for all critical alerts	P0
NT-04	Push notifications (iOS + Android)	P0
NT-05	Email daily digest + weekly performance report	P1
NT-06	Telegram two-way commands	P1
NT-07	Slack webhook for Institutional/Enterprise	P2
NT-08	Quiet market proactive communication	P1

3.6 Subscription & Billing

Req ID	Requirement	Priority
SB-01	14-day free trial (no credit card required)	P0
SB-02	Stripe billing for card payments (global)	P0
SB-03	Flutterwave for Africa bank transfer + mobile money	P0
SB-04	Paystack for South Africa EFT	P0
SB-05	M-Pesa via Flutterwave (Kenya)	P0
SB-06	Automated tier upgrade/downgrade within 60 seconds	P0
SB-07	Churn mitigation: P&L-based cancel retention message	P1
SB-08	Performance fee add-on (Pro+, Phase 5)	P2

3.7 Compliance & Security

Req ID	Requirement	Priority
CS-01	Platform never holds, moves, or controls user funds	P0
CS-02	API keys scoped to Read + Trade only (no Withdraw)	P0
CS-03	All credentials in GCP Secret Manager — never in DB	P0
CS-04	Per-user secret isolation	P0
CS-05	GDPR compliance (EU users)	P0
CS-06	POPIA compliance (South Africa)	P1
CS-07	NDPA compliance (Nigeria)	P1
CS-08	FSCA CASP registration	P1
CS-09	SEC Nigeria Digital Asset registration	P1
CS-10	Kenya VASP Act registration	P1

4. Out of Scope (v1.0)

- Platform custody of user funds
- Nigeria P2P fiat (NGN bank transfer) leg automation (legal review required)
- CEX-DEX arbitrage (Phase 5)
- Copy trading or social trading features
- Crypto-to-fiat off-ramp
- Leveraged positions beyond exchange-native perpetuals

